

CADE ROBERT ARMSTRONG

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EDUCATION

The University of Texas at Austin
Ph.D. in Aerospace Engineering

Jan 2026 - Present

The University of Texas at Austin
B.S. in Aerospace Engineering with Honors

Aug 2021 - May 2025
GPA 3.94/4

RESEARCH INTERESTS

My primary research interests lie at the intersection of **orbital mechanics**, **tour design**, **control theory**, **game theory**, **reinforcement learning**, and **multi-agent decision making**. My goal is to improve space mission efficiency through optimized decision-making and trajectory design. Consequently, I am interested in applications involving **multi-body gravitational systems**, **autonomous spacecraft cooperation**, **dynamic games**, and **learning-based approaches** for decision-making in complex environments.

PUBLICATIONS

C. Armstrong, R. Park, X. Liu, K. Gupta, and D. Fridovich-Keil, "Inferring Foresight-
edness in Dynamic Noncooperative Games," in *IEEE Robotics and Automation Letters*,
vol. 10, no. 12, pp. 13193-13200, Dec. 2025, doi: 10.1109/LRA.2025.3625521.

WORK EXPERIENCE

Center for Space Research, UT Austin

Research Engineering/Scientist Professional

September 2025 - December 2025

- Developing an algorithm to predict secular, long-period, and short-period orbital element variations under Earth's gravitational harmonics.
- Developing error reduction methods for inter-satellite range rate measurements on the GRACE-FO mission, mitigating tilt-to-length coupling effects in laser ranging interferometers.

RESEARCH EXPERIENCE

Control and Learning for Autonomous Robotics (CLeAR) Lab, UT Austin

Orbital Mechanics, Non-cooperative Game Theory - Research Internship *May 2023 - Dec 2025*

- Developed a novel **multi-agent** algorithm for **inferring player foresight** in collaboration with [Prof. David Fridovich-Keil](#), which uses **inverse dynamic game methods** and is capable of real-time performance.

- Developed a **reinforcement learning** framework to approximate the **infinite horizon value function** for **zero-sum games** in collaboration with Prof. Fridovich-Keil, enabling infinite-horizon control within finite-horizon solvers via **learned terminal costs**.
- Worked in collaboration with Prof. Fridovich-Keil and [Kushagra Gupta](#) to develop a novel **second-order** algorithm for finding **local Nash equilibria** in **zero-sum games**.
- Studied and debugged a **non-cooperative, dynamic game** solver that solves a **traveling salesman** problem in the context of **tour design** about the Jovian moons.

TEACHING-RELATED EXPERIENCE

Tutor Manager - Dept. of Aerospace Engineering *May 2024 - May 2025*

- Oversaw and ran tutoring program operations, including efficient student assignment based on student needs and tutor expertise.
- Supported tutors in teaching, scheduling, and performance.

Tutor - Dept. of Aerospace Engineering *December 2022 - May 2024*

- Assisted and taught fellow UT Austin students in various topics including dynamics and engineering computation problems.
- Taught and reinforced academic strategies to assist in the continued success of students.

RELEVANT COURSEWORK

Orbital Dynamics and Controls: Optimal Control Theory, Spacecraft Dynamics, Low Earth Orbits for Earth Observation, Attitude Dynamics, Dynamics, Linear System Analysis, Feedback Control Systems, Rocket Engineering Practicum II

Mathematics and Computer Programming: Theoretical Foundations for Reinforcement Learning, Calculus, Linear Algebra, Differential Equations, Intro. to Computer Programming, Engineering Computation, Elements of Software Design, Computational Methods for Structural Analysis

Miscellaneous: Flight Dynamics, Astrophysics, Propulsion

TECHNICAL SKILLS

Programming Languages

Python, Julia, MATLAB, Java, C++

Tools and Softwares

Simulink, Solidworks, Abaqus

Software Libraries

NumPy, Matplotlib, SciPy